

Distributed Systems Oral / Viva Study Material

1. Introduction to Distributed Systems

What is a Distributed System?

A distributed system is a collection of independent computers that appears to users as a single system.

Characteristics

- Resource sharing
- Concurrency
- Scalability
- Fault tolerance
- Transparency

Advantages

- Faster processing
- Resource sharing
- Reliability
- Scalability

Disadvantages

- Complex design
 - Security issues
 - Network dependency
 - Synchronization problems
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2. Remote Method Invocation (RMI)

What is RMI?

RMI (Remote Method Invocation) allows a Java object running on one JVM to invoke methods on an object running on another JVM.

Components of RMI

- Remote Interface
- Remote Object

- Stub
- RMI Registry
- Client
- Server

Steps in RMI

1. Create remote interface
2. Implement interface
3. Create server
4. Register object in RMI registry
5. Create client
6. Invoke remote method

Important Classes

- Remote
- RemoteException
- UnicastRemoteObject
- LocateRegistry

Advantages

- Easy distributed communication
- Object-oriented
- Built-in multithreading

Limitations

- Java-specific
- Slower than local calls

Viva Questions

What is RMI?

Java technology for invoking methods remotely.

What is Stub?

Client-side proxy object.

What is Skeleton?

Server-side dispatcher (older Java versions).

What is RemoteException?

Handles network communication failures.

How is multithreading achieved in RMI?

RMI automatically creates threads for multiple client requests.

3. CORBA

What is CORBA?

CORBA (Common Object Request Broker Architecture) is middleware used for communication between distributed objects.

What is ORB?

ORB (Object Request Broker) acts as a broker between client and server objects.

What is IDL?

IDL (Interface Definition Language) defines interfaces independent of programming language.

CORBA Components

- ORB
- IDL
- Naming Service
- Stub
- Skeleton
- Client
- Server

Steps in CORBA

1. Create IDL file
2. Generate Java files using idlj
3. Implement servant class
4. Create server
5. Start ORBD
6. Create client

Important Commands

Generate files

```
idlj -fall filename.idl
```

Compile

```
javac *.java PackageName/*.java
```

Start ORBD

```
orbd -ORBInitialPort 1060 -ORBInitialHost localhost
```

Why Java 8 is Required?

CORBA packages were removed after Java 8.

Viva Questions

What is ORBD?

CORBA naming service.

Why use ORBD?

To register and locate distributed objects.

What is object brokering?

Communication between distributed objects through middleware.

Difference between RMI and CORBA?

RMI is Java-specific. CORBA is language independent.

4. Berkeley Clock Synchronization Algorithm

Purpose

Synchronizes clocks in distributed systems.

Working

1. Master polls all clocks.
2. Calculates average time.
3. Sends adjustment values.
4. Clocks synchronize.

Advantages

- No UTC needed
- Reduces clock drift

Viva Questions

What is clock synchronization?

Maintaining same time across systems.

What is clock drift?

Difference in clocks over time.

5. Token Ring Mutual Exclusion

Purpose

Ensures only one process enters critical section.

Working

- Token circulates in ring.
- Only token holder enters critical section.
- Token passed to next process.

Advantages

- No deadlock
- Fair scheduling

Disadvantages

- Token loss stops system

Viva Questions

What is mutual exclusion?

Only one process accesses shared resource at a time.

What is token?

Permission message for critical section.

6. Bully Algorithm

Purpose

Leader election in distributed systems.

Working

- Highest process ID becomes coordinator.
- Election messages sent to higher-ID processes.
- Highest active process becomes leader.

Advantages

- Simple
- Fast in small systems

Disadvantages

- Large number of messages

Viva Questions

Why called Bully Algorithm?

Highest-ID process dominates others.

Message Complexity

$O(n^2)$

7. Ring Algorithm

Purpose

Leader election using logical ring.

Working

- Election message circulates in ring.
- Active processes add IDs.
- Highest ID becomes coordinator.

Advantages

- Simple implementation

Disadvantages

- Slower in large rings

Viva Questions

Message Complexity

$O(n)$

Which process becomes coordinator?

Highest active process.

8. OpenMP

What is OpenMP?

API for parallel programming in C/C++.

Important Directives

Parallel Region

```
#pragma omp parallel
```

Loop Distribution

```
#pragma omp for
```

Critical Section

```
#pragma omp critical
```

Advantages

- Faster execution
- Multi-core utilization

Viva Questions

What is a thread?

Lightweight execution unit.

What is parallel processing?

Executing tasks simultaneously.

What is a critical section?

Shared code section executed by one thread at a time.

9. Important Definitions

Distributed Computing

Using multiple computers to solve a problem.

Scalability

Ability to grow with increasing workload.

Fault Tolerance

System continues even after failures.

Transparency

Hides distribution details from users.

Middleware

Software layer between application and OS/network.

Concurrency

Multiple processes executing simultaneously.

10. Frequently Asked Oral Questions

Difference between Process and Thread

Process:

- Independent execution unit
- Separate memory

Thread:

- Lightweight
 - Shared memory
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Difference between Parallel and Distributed Computing

Parallel Computing:

- Multiple processors in same machine

Distributed Computing:

- Multiple networked computers
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Difference between RMI and Socket Programming

RMI:

- Object-oriented
- Easier

Socket:

- Low-level communication
 - Faster
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Difference between CORBA and RMI

CORBA:

- Language independent

RMI:

- Java-specific
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11. Important Commands Summary

RMI Compilation

```
javac *.java
```

RMI Server

```
java RMIServer
```

RMI Client

```
java RMIClient
```

CORBA IDL Generation

```
idlj -fall filename.idl
```

CORBA Compile

```
javac *.java PackageName/*.java
```

Start ORBD

```
orbd -ORBInitialPort 1060 -ORBInitialHost localhost
```

Run Server

```
java -cp . ServerName -ORBInitialPort 1060 -ORBInitialHost localhost
```

Run Client

```
java -cp . ClientName -ORBInitialPort 1060 -ORBInitialHost localhost
```

12. Tips for Oral/Viva

- Explain program flow clearly.
- Know purpose of every file.

- Remember architecture diagrams.
 - Learn definitions.
 - Explain advantages and disadvantages.
 - Understand error messages.
 - Practice execution steps.
 - Know difference between algorithms.
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13. Common Errors and Fixes

RMI Error

Port already in use

Use another port.

CORBA Error

Connection refused

ORBD not running.

CORBA Error

org.omg packages missing

Use Java 8.

Class Not Found

Use:

```
java -cp . ClassName
```

14. Architecture Diagrams

RMI

Client ---> RMI Registry ---> Remote Object(Server)

CORBA

Client ---> ORB ---> Remote Object(Server)

Token Ring

P1 -> P2 -> P3 -> P4 -> P1

Ring Election

Election Message circulates in ring